PSEB / ATOMCAMP DEPLOY AI BOOTCAMP — TECHNICAL REPORT

Client-Side Attacks, Payloads & Post-Exploitation Analysis

Author: Zahid Ullah | **Instructor:** Waseem Abbas Khan | **Date:** November 9, 2026 | **Module:** S08

Target Environment:	Isolated VMware Enterprise Lab (pfSense 2.6.0 Routed)
Attacker Host:	Kali Linux 2024.x (10.50.20.10 / VMnet3 Offensive Security)
Target Workstation:	Windows 10 Pro 64-bit (10.50.10.21 / VMnet2 Enterprise LAN)
Regulatory Authority:	Authorized Laboratory Testing under Pakistan PECA 2016 Permitted Scope
Assessment Focus:	Malware Taxonomy, Staged Payloads, MIC / UAC, Named Pipe Impersonation & Career Path

1. Executive Summary & Scope of Assessment

Modern enterprise perimeter defenses actively filter and reject unsolicited inbound network connections. Consequently, adversaries and red team practitioners rely heavily upon client-side operations where code execution is initiated from within the internal protected network. This technical report provides a rigorous academic and empirical evaluation of client-side payloads, malware classification, user interaction deciders, Windows access token dynamics, and post-exploitation privilege elevation within an isolated dual-homed virtual laboratory.

> **LABORATORY INTEGRITY & AUTHORIZATION STATEMENT:** All testing and payload executions were conducted in a strictly isolated, dual-subnet virtual environment with no external routable connectivity. All actions were authorized for research and training under Pakistan's Prevention of Electronic Crimes Act (PECA 2016).

2. Task 1: Comprehensive Malware Taxonomy & Behavioral Analysis

Modern defensive architectures require granular knowledge of malware operational behaviors, propagation mechanisms, and defensive countermeasures. The matrix below presents the 9 primary malware classifications with real-world exemplars and mitigations.

Family	Spread Vector	Real-World Exemplar	Mechanism of Action	Primary Defense
Virus	File transfer, host executable infection	CIH (Chernobyl), Stuxnet LNK	Appends malicious code to benign host executables; triggers on host run.	Antivirus heuristics, File Integrity Monitoring (FIM).
Worm	Network flaws, autonomous scan	WannaCry (MS17-010), Conficker	Self-replicates across subnets utilizing unpatched network vulnerabilities.	Network micro-segmentation, rapid vulnerability patching.
Trojan	Social engineering, fake utilities	Emotet, Zeus Banking Trojan	Masquerades as authentic software while deploying stealth background payloads.	Application Control (WDAC/AppLocker), EDR heuristics.
Ransomware	Phishing, RDP compromise	LockBit 3.0, Conti, Ryuk	Encrypts files via asymmetric keys, demanding payment for private decryptor.	Immutable offline backups, canary file detection, least privilege.
Spyware	Bundled installers, rogue plugins	Pegasus, FinFisher, DarkComet	Harvests keystrokes, webcam feeds, credentials, and exfiltrates silently.	Egress filtering, browser policy enforcement, EDR telemetry.

Adware	Deceptive bundle installers	Superfish, Fireball	Hijacks browser search engines and forces unwanted pop-up advertisements.	Secure Web Gateways, ad-block policies, enterprise browser control.
Rootkit	Kernel exploits, signed drivers	TDSS / Alureon, Sony BMG	Hooks kernel syscalls and SSDT to hide files, network sockets, and processes.	UEFI Secure Boot, Driver Signature Enforcement, VBS.
Botnet	Credential spray, automated bots	Mirai, Necurs, Trickbot	Subjugates infected hosts into an orchestrated C2 network for DDoS or spam.	Egress DNS sinkholing, Network Intrusion Detection (NIDS).
Fileless	RAM injection, LOLBins (PowerShell)	Astaroth, Poweliks, Cobalt Strike	Executes script payloads directly in volatile memory without dropping PE files on disk.	AMSI runtime scanning, PowerShell CLM, Script Block Logging.

2.1 The Decisive Role of User Interaction in Client-Side Attacks

In modern networks protected by stateful inspection and NAT, unsolicited remote exploitation from the outside is largely blocked. User interaction serves as the critical security bypass. When an authorized user downloads or opens a payload, three decisive shifts occur: **(1) Firewall Egress Traversal:** The connection originates from inside, traversing perimeter firewalls; **(2) Security Token Inheritance:** The payload executes within the user's active session, inheriting their access privileges; **(3) Trust Exploitation:** The user's action diminishes rule-based heuristic alerts by mimicking standard user behavior.

2.2 Deep-Dive: Fileless Malware & Living-off-the-Land Binaries (LOLBins)

To evade signature-based antivirus solutions, fileless malware leverages signed Windows binaries already present on the system (LOLBins):

- **PowerShell (powershell.exe):** Downloads and reflects payloads into memory without writing to disk.
- **Certutil (certutil.exe):** Abused via *-urlcache -split -f* to download and decode remote payloads.
- **MSHTA (mshta.exe):** Executes remote VBScript/JScript files directly from the web, bypassing browser execution checks.
- **BITSAdmin (bitsadmin.exe):** Initiates asynchronous, throttling-resilient background downloads mimicking OS updates.

Defensive Mitigation: Requires runtime behavioral inspection: Antimalware Scan Interface (AMSI), PowerShell Script Block Logging (Event ID 4104), and Constrained Language Mode (CLM) enforced via WDAC.

3. Task 2: Payload Engineering, Execution Telemetry & Privilege Analysis

3.1 Comparative Analysis: Staged vs. Inline (Non-Staged) Payloads

Metasploit denotes payload architecture through forward slash notation (*windows/meterpreter/reverse_tcp* = staged) versus underscore notation (*windows/meterpreter_reverse_tcp* = inline). A staged payload generates an ultra-lightweight initial binary (stager: 355 bytes shellcode in a 7,168-byte PE). The stager's sole purpose is to initialize Winsock, connect back to the listener, allocate memory via *VirtualAlloc()*, and download the full 203,451-byte reflective DLL stage. An inline payload packages the entire reflective DLL (~200+ KB) inside the initial binary, eliminating secondary network transmission but increasing static binary footprint.

3.2 Laboratory Setup, Delivery & Listener Activation

Testing was conducted across two segmented subnets: Kali Linux (10.50.20.10) and Windows 10 (10.50.10.21) separated by pfSense. While Kali cannot ping Windows 10 due to Windows Defender Firewall dropping inbound ICMP, Windows 10 successfully reaches Kali with round-trip latency <1 ms (TTL=63).

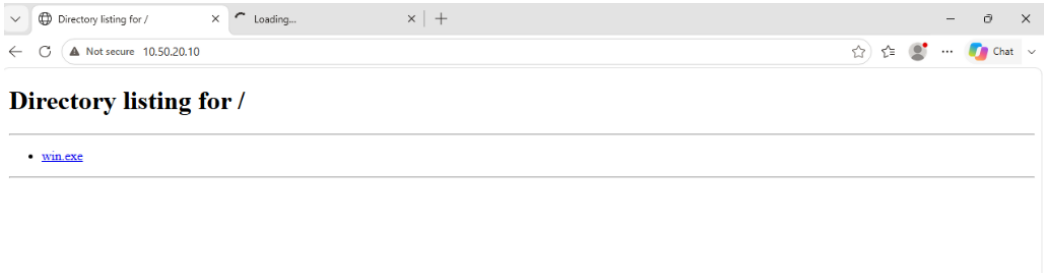


Figure 2.1: *Payload delivery interface hosted via Python HTTP on Kali accessed from Windows 10 Edge.*

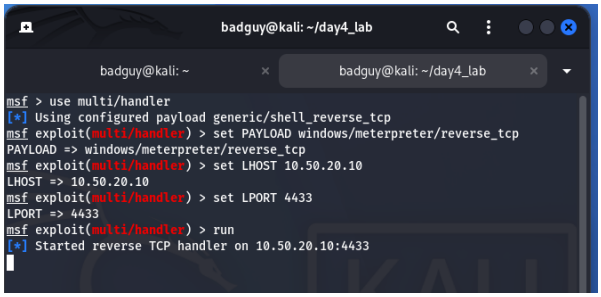


Figure 2.2: *Metasploit multi/handler configured on 10.50.20.10:4433 listening for incoming callback.*

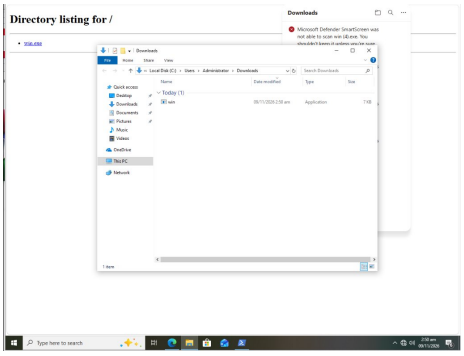


Figure 2.3: *Target filesystem verification confirming download of win.exe in Downloads directory.*

3.3 Target Execution & Active Session Monitoring

Upon simulated execution on Windows 10, Task Manager confirmed the creation of the 32-bit container win.exe (consuming 1.0 MB RAM). The process immediately initiated a reverse TCP connection to port 4433, prompting Metasploit to transmit the 203,451-byte reflective DLL stage and establish Session 1.

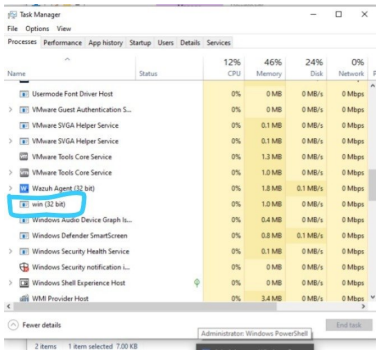


Figure 2.4: *Windows Task Manager confirming active execution of 32-bit win.exe container.*

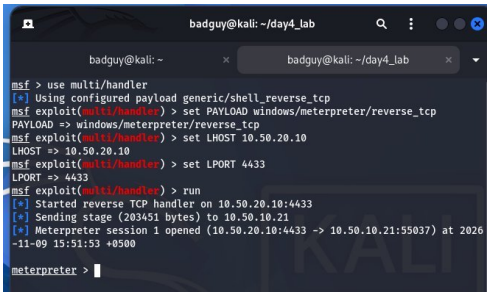


Figure 2.5: Metasploit console logging stage delivery (203,451 bytes) and Meterpreter Session 1 opened.

3.4 Access Token Mechanics & Named Pipe Impersonation (Technique 1)

Windows implements Mandatory Integrity Control (MIC) and Access Tokens. Administrative accounts operate under filtered Medium-integrity tokens by default under User Account Control (UAC). In our verified lab session, the process ran under the domain administrator context (*SEC\Administrator*). Upon running *getsystem*, Meterpreter leveraged **Technique 1 (Named Pipe Impersonation)**:

1. Meterpreter creates a local Windows Named Pipe (\\.\pipe\...).
2. As an administrator, it creates and starts a temporary service running under *NT AUTHORITY\SYSTEM*.
3. The SYSTEM service connects to the listening named pipe.
4. Meterpreter calls the Windows API *ImpersonateNamedPipeClient()*, stealing the connecting client's SYSTEM access token and assuming full OS control.

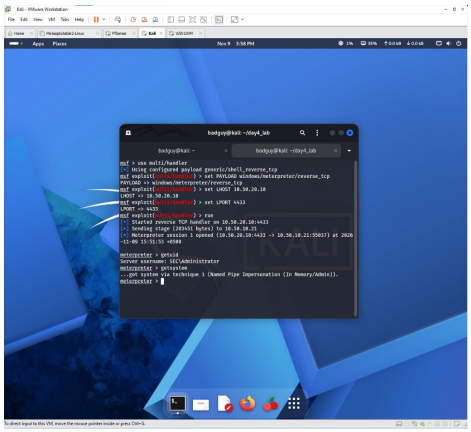


Figure 2.6: Verification of initial user identity (SEC\Administrator) and elevation to NT AUTHORITY\SYSTEM via Named Pipe Impersonation.

3.5 Advanced Delivery Concepts & Legal Framework

• **WinRAR SFX Binding:** Malicious executables are bundled with legitimate utilities (e.g., Rufus, PDF readers). The SFX script silently extracts both files to %TEMP%, executing the decoy utility visually while running the payload invisibly in the background.
• **WAN Ingress Tunneling (LocalToNet/Ngrok):** Provides a public TCP relay address, enabling reverse TCP callbacks to traverse carrier-grade NAT (CGNAT) without exposing the attacker's physical IP address.
• **Legal Governance (Pakistan PECA 2016):** Section 3 (Unauthorized access to systems/data), Section 4 (Unauthorized copying/transmission of data), and Section 5 (Interference with data/systems) establish clear criminal penalties. All activities must be bound by signed Rules of Engagement (RoE) and asset owner authorization.

4. Task 3: Professional Career Statement & 90-Day Execution Roadmap

As a graduate of the PSEB / Atomcamp DEPLOY AI Cybersecurity Bootcamp, my career target is to serve as a **Security Operations Center (SOC) Analyst / Junior Penetration Tester**. By combining offensive insight into malware staging and access token mechanics with defensive detection engineering, I aim to protect enterprise infrastructures from sophisticated threats.

- **Industry Credential Pathway:** CompTIA Security+ (Months 1-2), Blue Team Level 1 / BTL1 (Months 3-4), and Practical Network Penetration Tester / PNPT (Months 5-6).

• **GitHub Portfolio Project 1 (Days 1-30):** Enterprise Multi-Subnet Virtual Range & pfSense Traffic Routing Documentation with Wazuh EDR integration.

• **GitHub Portfolio Project 2 (Days 31-60):** LOLBins Threat Hunting & Sigma Detection Repository (auditing Certutil, MSHTA, and PowerShell memory execution).

• **GitHub Portfolio Project 3 (Days 61-90):** Automated Windows Endpoint CIS Hardening & Audit Script (evaluating UAC levels, Script Block Logging, and disabling legacy protocols).

5. Evidence Repository Index

Artifact Identifier	Type	Description & Phase
01_network_verification.txt	Text Log	Bidirectional reachability across pfSense between Kali and Windows 10.
02_msfvenom_payload_gen.txt	Text Log	Stager payload generation parameters and binary sizing metrics.
02_payload_delivery_http.png	Visual Capture	Target accessing Python HTTP web server for payload acquisition.
03_msf_handler_listen.png / .txt	Visual / Log	Metasploit multi/handler listener configuration on port 4433.
04_payload_downloaded_windows.png	Visual Capture	Target filesystem verification in Windows 10 Downloads folder.
05_task_manager_execution.png	Visual Capture	Target Task Manager confirming active win.exe (32-bit) process.
06_meterpreter_session_established.png	Visual Capture	Stage delivery (203,451 bytes) and Session 1 opened on Kali.
07_privilege_elevation_system.png / .txt	Visual / Log	Verification of SEC\Administrator session and Named Pipe Impersonation to SYSTEM.